

Total No. of Questions : 8]

SEAT No. :

P3195

[Total No. of Pages : 2

[5670]-297

B.E. (Computer)

DATA MINING TECHNIQUES AND APPLICATIONS

(2012 Pattern) (Semester - I) (Elective - I)

Time : 2½ Hours]

[Max. Marks :70

Instructions to the candidates:

- 1) Answer questions 1 or Q2, Q.3 or Q.4, Q.5 or Q6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

- Q1)** a) Explain data integration, data transformation and data reduction in short. [6]
- b) Write and explain FP Growth Algorithm for calculating frequent item sets. How it is better than Apriori algorithm. [8]
- c) Describe performance metrics for evaluating classifiers with equations. [6]

OR

- Q2)** a) Explain following procedures for attribute subset selection: [8]
- i) Stepwise forward selection.
 - ii) Stepwise backward elimination.
- b) Explain the following in short: [8]
- i) Regression Classification Algorithm.
 - ii) ID3 Decision trees algorithm.
- c) Write a short note on constraint based association rule mining. [4]
- Q3)** a) Explain cluster analysis. Write and explain K-Means Clustering algorithm. [6]
- b) What is Hierarchical Clustering? Explain Agglomerative and Divisive Hierarchical Clustering with neat diagrams. [5]
- c) Explain different distance measures in brief. [6]

OR

P.T.O.

- Q4)** a) Write typical requirements of clustering in data mining. Describe any five. [5]
 b) Write and explain PAM Clustering algorithm. [8]
 c) Using the two given objects represented by the tuples (22, 1, 42, 10) and (20, 0, 36, 8), compute the following: [4]
 i) Euclidean distance
 ii) Manhattan distance

- Q5)** a) Explain the following basic measures for text retrieval: [6]
 i) Precision
 ii) Recall
 iii) F score
 b) Write and explain Hyperlink-Induced Topic (HITS) algorithm. [6]
 c) Which methods are used for Dimensionality Reduction of Text in text mining? Describe any one in brief. [5]

OR

- Q6)** a) Describe the following terms in brief: Feature vector, Term frequency, Bag of words and Inverse document frequency. [8]
 b) Write a short note on: [6]
 i) Web content mining
 ii) Web usage mining.
 c) What is document ranking? [3]

- Q7)** a) Write a short note on. [10]
 i) Reinforcement learning
 ii) Wholistic Learning
 b) Describe multi-perspective learning along with diagram. [6]

OR

- Q8)** a) Explain the following with respect to reinforcement learning: Intelligent agent and environment, Learning agents, Rewards and Adaptive learning. [8]
 b) Describe the advanced techniques for big data mining. [8]

